

tf.compat.v1.nn.embedding_lookup_sparse

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/embedding_lookup_sparse

Looks up embeddings for the given ids and weights from a list of tensors.

Function Signatures

```
tf.compat.v1.nn.embedding_lookup_sparse( params, sp_ids,
sp_weights, partition_strategy='mod', name=None,
combiner=None, max_norm=None, allow_fast_lookup=False )
```

Code Examples

```
tf.compat.v1.nn.embedding_lookup_sparse(
params,
sp_ids,
sp_weights,
partition_strategy='mod',
name=None,
combiner=None,
max_norm=None,
allow_fast_lookup=False
)
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    params,  
    sp_ids,  
    sp_weights,  
    partition_strategy='mod',  
    name=None,  
    combiner=None,  
    max_norm=None,  
    allow_fast_lookup=False  
)
```

```
[0, 0]: id 1, weight 2.0  
[0, 1]: id 3, weight 0.5  
[1, 0]: id 0, weight 1.0  
[2, 3]: id 1, weight 3.0
```

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[0, 0]: id 1, weight 2.0  
[0, 1]: id 3, weight 0.5  
[1, 0]: id 0, weight 1.0  
[2, 3]: id 1, weight 3.0
```

```
output[0, :] = (params[1, :] * 2.0 + params[3, :] * 0.5) / (2.0 +  
0.5)  
output[1, :] = (params[0, :] * 1.0) / 1.0  
output[2, :] = (params[1, :] * 3.0) / 3.0
```

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output[1, :] = (params[0, :] * 1.0) / 1.0  
output[2, :] = (params[1, :] * 3.0) / 3.0
```

Documentation

Looks up embeddings for the given ids and weights from a list of tensors.

This op assumes that there is at least one id for each row in the dense tensor represented by `sp_ids` (i.e. there are no rows with empty features), and that all the indices of `sp_ids` are in canonical row-major order.

`sp_ids` and `sp_weights` (if not `None`) are `SparseTensors` or `RaggedTensors` with rank of 2. For `SparseTensors` with left-aligned non-zero entries which can be described as `RaggedTensors`, use of `RaggedTensors` can yield higher performance.

It also assumes that all id values lie in the range $[0, p_0)$, where p_0 is the sum of the size of params along dimension 0.

Returns

In other words, if

`shape(combined params) = [p0, p1, ..., pm]`

`shape(sp_ids) = shape(sp_weights) = [d0, d1]`

`shape(output) = [d0, p1, ..., pm]`.

For instance, if `params` is a 10x20 matrix, and `sp_ids` / `sp_weights` are with `combiner="mean"`, then the output will be a 3x20 matrix where

Raises

